Ahnaf Shahriar

Email | LinkedIn | Github

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Computer Engineering — *Sept. 2021 – Apr. 2026*

- **GPA**: 3.7
- **Relevant Courses**: Computer Architecture, Distributed Computing, ML for Chip Design, Robot Dynamics & Control, Compilers, Real-Time Operating Systems

Experience

Undergraduate Research Assistant — May. 2025 – Present
Wireless Sensors and Devices Lab — *Waterloo, ON*

- **BLE development**: Designing and firmware for *Extended Advertisement* for ultra low power mesh networks
- **Wireless Gateway**: Implementing 5G provisioning for gateway devices with Satellite connectivity.

Wireless Firmware Developer — Jan. 2025 – Apr. 2025
ON Semiconductor Corporation — *Waterloo, ON*

- **SDK Development**: Integrated *I3C* drivers and *Bluetooth Low Energy* profile services into sample code.
- **Wireless Connectivity**: Developed and integrated cutting-edge *LE Audio, HFP, and A2DP* functionality into an all-in-one Application.
- **Wireless Manager** Designed a modular state machine for bluetooth, improving power efficiency by *25%*
- **Test Automation**: Automated SDK release test cases, reducing testing time by up to *80%* for samples

IC Design and Verification Intern — May 2023 – Aug. 2023, Jan. 2024 – May 2024
NXP Semiconductors Canada — *Kanata, ON*

- **IP Design**: Designed multiple IP blocks including *proprietery ECC IP* for dataplane processing chips.
- **Timing Analysis**: Utilized *Synopsys Design Compiler* and improved Parsing IP Clock Frequency by *20%.*
- **Functional Testing**: Designed SystemVerilog End-to-End functional tests in simulating *10GBps+* traffic for IP.
- **Unit Test Planning**: Created Simulation scenarios for testing High speed Dataplane Processing features.

Embedded Software Engineering Intern — Sept. 2022 – Dec. 2022
Synapse Product Development — *Seattle, WA*

- **Prototyping**: Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs**: Designd custom lab automation APIs for equipment from *Agilent, Keysight, NI, Tektronik.*
- **Automation**: Streamlined testing and in house procedures using *Python* and *Bash.*
- **Driver Development**: Wrote low-level *I2C and UART* drivers for in-house PCB test platforms on Zephyr RTOS.

Firmware Developer — Jan. 2022 – April 2022
Ford Motor Company of Canada — *Remote*

- **Unity/Cmock Test framework**: Lead developer for optimization for unit testing, achieving up to *30% faster* runtime while using *50%* less manually written test cases.
- **Automation**: Improved *Jenkins CI/CD* pipelines to support unit testing automation using *Python*
- **Embedded Trace Debugging**: Built high speed Parsing tools to parse through CAN and Serial messages exceeding *10 million+* lines.

Projects

LC-3 Emulator: A C emulator for an educational ISA. Improves by 25% on research paper using *Python* data logging.
Real Time Executable: A RTOS implementation in STM32 capable of Pre-emptive task switching and its own Malloc
Stereo System: An embedded C implementation of a stereo playback system. Created with Quartus on Artix FPGA.
VHDL Compiler: A Java Compiler for creating VHDL circuits. Using a boolean intermediate representation

Technical Skills

Languages: C/C++, Java, Python, Tcl, Bash scripting, ASM, VHDL, SystemVerilog/Verilog
Tools: Keil, Quartus, Git, Linux, Qemu, LLDB/GDB , Docker, WireShark, UVM, Matlab
Hardware: Oscilloscopes, Logic Analyzer, Multimeters, Spectrum Analyzer
Protocols: TCP/IP, JTAG, Serial, Ethernet, CAN/CAN-FD, LIN, AXI